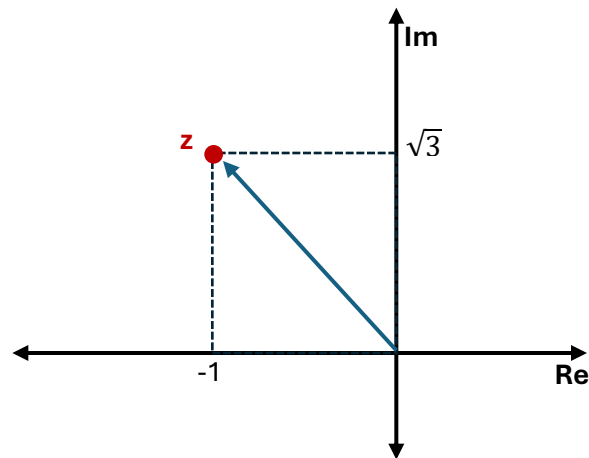


**Homework Set 1. Math Prerequisites, Dynamic Modeling, and Simulation (Solutions)****Question 1****(a)**

$$z = -1 + \sqrt{3}i$$

**(b)**

$$z = -1 + \sqrt{3}i$$

- Modulus:

$$|z| = \sqrt{1 + 3} = 2$$

- Argument

For  $x < 0$ , and  $y \geq 0$ , the argument of a complex number  $z = x + iy$  can be calculated by:

$$\angle z = \tan^{-1}\left(\frac{y}{x}\right) + \Pi.$$

$$\text{Hence, } \angle z = \tan^{-1}(-\sqrt{3}) + \Pi = \frac{2\Pi}{3} \text{ radians}$$

**(c)**

Let :

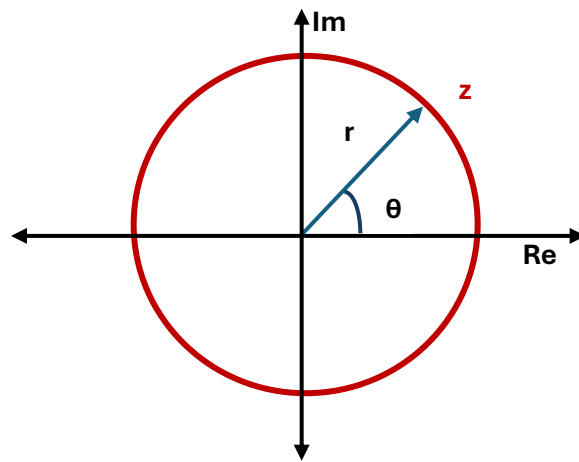
$$z = e^{i\theta} = \cos \theta + i \sin \theta.$$

From part b,

$$|z| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1.$$

(d)

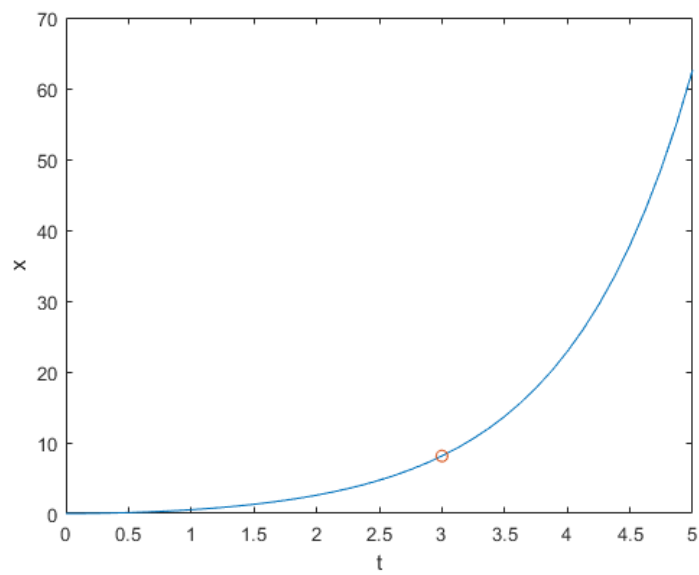
$$z = re^{i\theta} \quad (r > 0 \text{ and } \theta \text{ are both real})$$



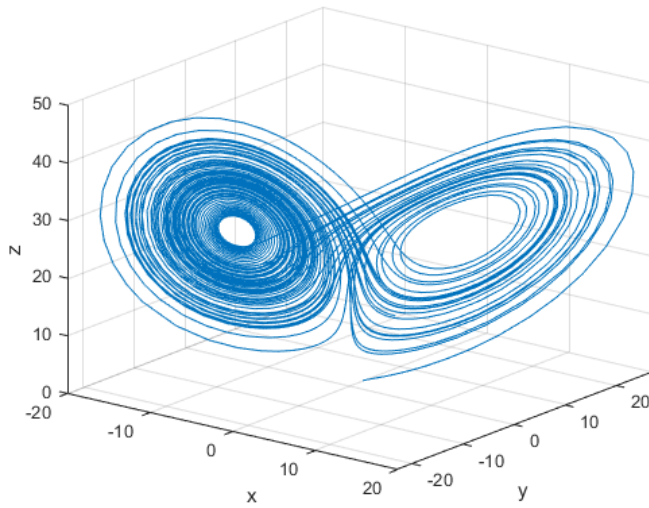
## Question 2

(a)

$$x(t = 3) \approx 8.1213$$



(b)



### MATLAB Code

```
% Part (a)
% beginning and end t-values
t_range = [0,5];
% initial condition
x0 = 0;

% solve ODE with ode45
[t_sol,x_sol] = ode45(@my_ode1, t_range, x0);
% find x(3)
x_3 = interp1(t_sol, x_sol, 3);
% plot solution
plot(t_sol, x_sol, '-', 3, x_3, 'o');
xlabel('t');
ylabel('x');

% define ODE function
function dxdt = my_ode1(t, x)
dxdt = sqrt(t^2 + x^2);
end
% -----
% Part (b)
% beginning and end t-values
t_range = [0,60];
% initial condition
IC = [1; 1; 1];

% solve ODE with ode45
[t_sol,Vec_sol] = ode45(@my_ode2, t_range, [1; 1; 1]);
[x_sol, y_sol, z_sol] = deal(Vec_sol(:,1), Vec_sol(:,2), Vec_sol(:,3));
% plot solution
```

```

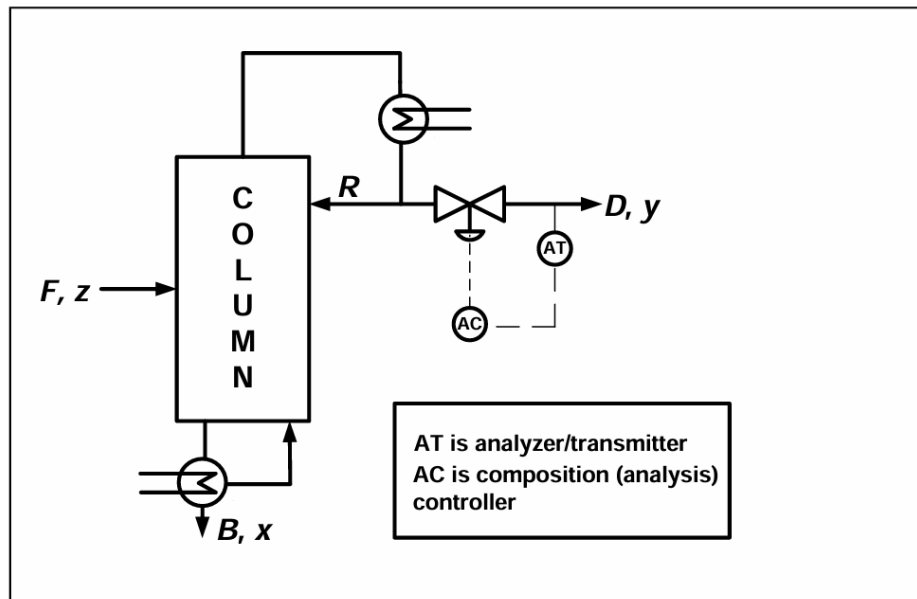
plot3(x_sol, y_sol, z_sol, '-');
grid("on");
xlabel('x');
ylabel('y');
zlabel('z');

% define ODE function
function dVecdt = my_ode2(t, variables)
[x, y, z] = deal(variables(1), variables(2), variables(3));
[s, r, b] = deal(10, 28, 8/3);
dxdt = s*(y-x);
dydt = x*(r-z) - y;
dzdt = x*y - b*z;
dVecdt = [dxdt; dydt; dzdt];
end

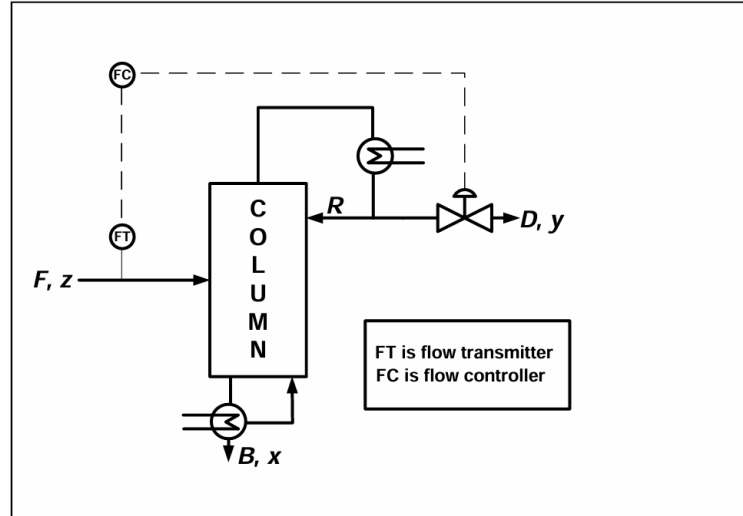
```

### Question 3

- (a) Feedback Control:  
Measure variable:  $y$   
Manipulated variable:  $D$ ,  $R$ , or  $B$



- (b) Feedforward Control:  
Measured variable:  $F$   
Manipulated variable:  $D$ ,  $R$ , or  $B$



#### Question 4

(a)

Mass Balances:

$$\rho A_1 \frac{dh_1}{dt} = w_1 - w_2 - w_3 \quad (1)$$

$$\rho A_2 \frac{dh_2}{dt} = w_2 \quad (2)$$

Flow relations:

Let  $P_1$  be the pressure at the bottom of tank 1.

Let  $P_2$  be the pressure at the bottom of tank 2.

Let  $P_a$  be the ambient pressure.

Then

$$w_2 = \frac{P_1 - P_2}{R_2} = \frac{\rho g}{g_c R_2} (h_1 - h_2) \quad (3)$$

$$w_3 = \frac{P_1 - P_a}{R_3} = \frac{\rho g}{g_c R_3} h_1 \quad (4)$$

(b)

Seven parameters:  $\rho, A_1, A_2, g, g_c, R_2, R_3$

Five variables :  $h_1, h_2, w_1, w_2, w_3$

Four equations

Thus  $N_F = 5 - 4 = 1$

1 input =  $w_1$  (specified function of time)

4 outputs =  $h_1, h_2, w_2, w_3$

### Question 5

Assume the feed contains only A and B, and no C.

- Component balances in the reactor:

$$\frac{dn_A}{dt} = q_i c_{Ai} - q c_A - V r_1$$

$$\frac{dn_B}{dt} = q_i c_{Bi} - q c_B + V r_1 - V r_2$$

$$\frac{dn_C}{dt} = -q c_C + V r_2$$

Plug in  $n_A = V c_A, n_B = V c_B, n_C = V c_C, r_1 = k_1 \exp\left(-\frac{E_1}{RT}\right) c_A, r_2 = k_2 \exp\left(-\frac{E_2}{RT}\right) c_B$

$$V \frac{dc_A}{dt} = q_i c_{Ai} - q c_A - V k_1 \exp\left(-\frac{E_1}{RT}\right) c_A$$

$$V \frac{dc_B}{dt} = q_i c_{Bi} - q c_B + V \left[ k_1 \exp\left(-\frac{E_1}{RT}\right) c_A - k_2 \exp\left(-\frac{E_2}{RT}\right) c_B \right]$$

$$V \frac{dc_C}{dt} = -q c_C + V k_2 \exp\left(-\frac{E_2}{RT}\right) c_B$$

- Mass balance in the jacket:

Since the volume in the jacket does not change, we can write:

$$q_{ci} = q_c.$$

- Energy balance for the reactor:

Assuming the molar heat capacity for A, B, C are  $C_{P,A}, C_{P,B}, C_{P,C}$ , which do not change with temperature.

The general energy balance equation:

$$\frac{dH}{dt} = \dot{H}_{in} - \dot{H}_{out} + \dot{Q}$$

Also, knowing using the relation between enthalpy of material A, B, and C to remove  $c_C C_{P,C}$  on the right-hand-side:

$$\begin{aligned}\hat{H}_B &= \hat{H}_A + \Delta H_1 = C_{P,A}T + \Delta H_1 \\ \hat{H}_C &= \hat{H}_A + \Delta H_1 + \Delta H_2 = C_{P,A}T + \Delta H_1 + \Delta H_2 = C_{P,B}T + \Delta H_2\end{aligned}$$

The total energy in the reactor:

$$H = V[c_A C_P + c_B(C_{P,A}T + \Delta H_1) + c_C(C_{P,A}T + \Delta H_1 + \Delta H_2)]T$$

$\dot{H}_{in}$  and  $\dot{H}_{out}$  are as follows:

$$\dot{H}_{in} = q_i(c_{Ai}C_{P,A} + c_{Bi}C_{P,B})T_i$$

$$\dot{H}_{out} = q[c_A C_{P,A}T + c_B(C_{P,A}T + \Delta H_1) + c_C(C_{P,A}T + \Delta H_1 + \Delta H_2)]$$

The  $\dot{Q}$  = heat exchange with jacket:

$$\dot{Q} = -UA(T - T_c)$$

Plug in these equations to get a general form of energy balance:

$$\begin{aligned}\frac{d[V(c_A C_{P,A} + c_B C_{P,B} + c_C C_{P,C})T]}{dt} = \\ q_i(c_{Ai}C_{P,A} + c_{Bi}C_{P,B})T_i - q(c_A C_{P,A} + c_B C_{P,B} + c_C C_{P,C})T - UA(T - T_c)\end{aligned}$$

For further simplification, first expand the derivation on the left-hand-side, and then replace  $c_C$  by

$$c_C = c_{Ai} + c_{Bi} - c_A - c_B$$

After plugging in equations of species balances, the final simplified form is:

$$\begin{aligned}V(c_A C_{P,A} + c_B C_{P,B} + c_C C_{P,C})\frac{dT}{dt} = q_i(c_{Ai}C_{P,A} + c_{Bi}C_{P,B})(T_i - T) - UA(T - T_c) + \\ (-\Delta H_1)Vk_1 \exp\left(-\frac{E_1}{RT}\right)c_A + (-\Delta H_2)Vk_2 \exp\left(-\frac{E_2}{RT}\right)c_B.\end{aligned}$$

It is also appropriate to use the specific heat to replace the molar heat capacity by:  $C_{P,A}(\frac{kJ}{mol \cdot K}) = M_A S_A(\frac{kJ}{kg \cdot K})$ , where  $M_A$  is the molecular weight for A. Similarly, this applies to B and C:

$$\begin{aligned}V(c_A M_A S_A + c_B M_B S_B + c_C M_C S_C)\frac{dT}{dt} = q_i(c_{Ai}M_A S_A + c_{Bi}M_B S_B)(T_i - T) - UA(T - T_c) + \\ (-\Delta H_1)Vk_1 \exp\left(-\frac{E_1}{RT}\right)c_A + (-\Delta H_2)Vk_2 \exp\left(-\frac{E_2}{RT}\right)c_B\end{aligned}$$

- Energy balance for the jacket:

$$V_j c_j C_P \frac{dT_c}{dt} = q_{ci} c_j C_P (T_{ci} - T_c) + UA(T - T_c) \text{ or}$$

$$\rho_j V_j S_j \frac{dT_c}{dt} = \rho_j q_{ci} S_j (T_{ci} - T_c) + UA(T - T_c)$$

Where

$\rho_j$  is the density of the coolant.

$S_j$  is the specific heat of the coolant.

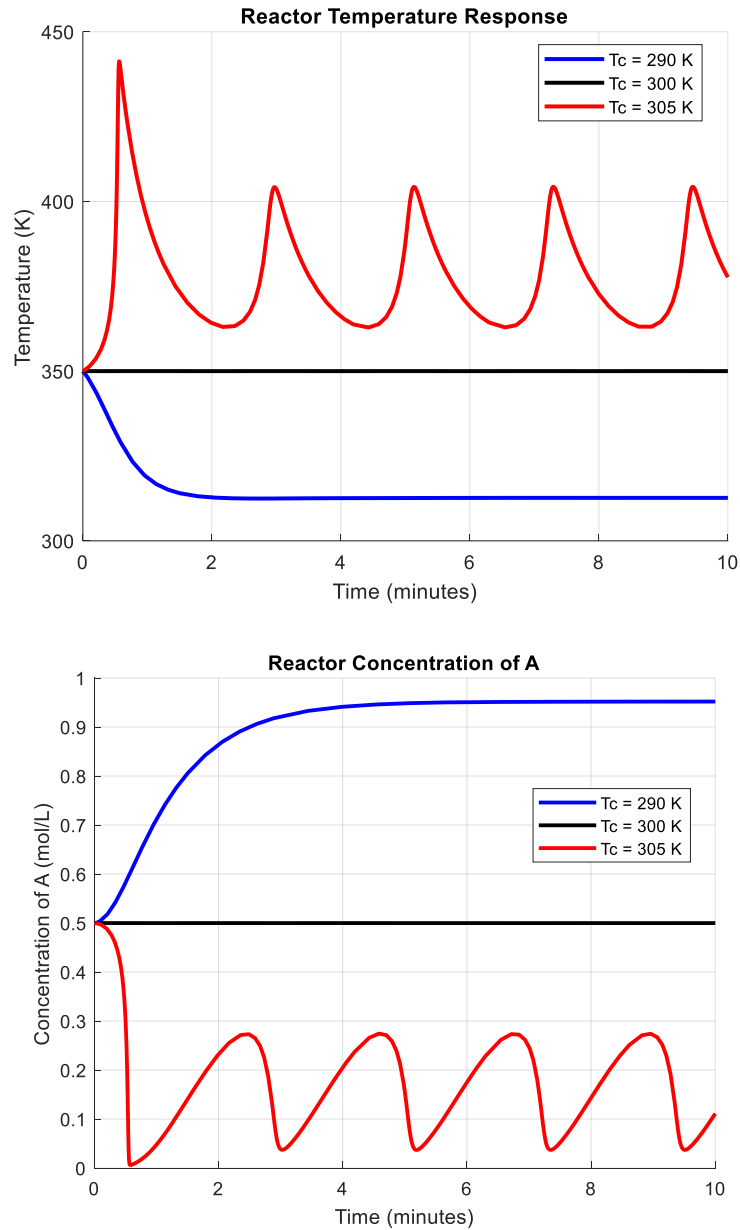
$V$  is the volume of the jacket.

### Question 6

| Category:          | Example Type:             | Mechanism:                                                                                                                                                                                                                              | Application:                                                                                                                                                                                                      |
|--------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Liquid Column      | Manometer                 | A U-shaped tube where one end is open to the atmosphere and the other is influenced by the fluid being measured. A column of liquid (e.g. water) is measured by its height difference on either side of the U-bend.                     | Measures the gauge pressure of a fluid in a pipe or vessel. Typically limited on fluid compatibility with the column liquid fluid.                                                                                |
| Elastic Element    | Bourdon Tube              | A thin tube bent in an arc and sealed at one end, sees the fluid being measured. The pressure of the fluid straightens the tube which causes a rotation in a needle to represent a pressure measurement.                                | Measures the gauge pressure of a fluid in a pipe or vessel. Comes in various spans of pressure measurement ranges. Very common instrument in industry.                                                            |
| Electrical Sensing | Piezoelectric Transducers | Utilizes piezoelectric material where a strain caused by a change in pressure causes a change in electric charge that can be measured as a change in voltage. The voltage change can be related as proportional to the pressure change. | Linear deformation over a relatively large span and immunity to EMF allows Piezoelectric Transducers to be used in harsh environments. Capable in micro-scale sensors as a micro-electromechanical system (MEMS). |



### Question 7



### Discussion:

The plots above depict the reactor's nonlinear dynamic behavior, as the cooling jacket temperature is changed. At lower temperatures, such as 290 K, the reactor maintains a relatively stable operation, illustrating how effective heat removal helps sustain system equilibrium. However, at higher temperatures, such as 305 K, significant oscillations and instability occur due to inadequate cooling, leading to temperature spikes and corresponding fluctuations in reactant A concentration.

## MATLAB Code

```
% Parameters
V = 100;           % Reactor volume in L
q = 100;           % Flow rate in L/min
rho = 1000;        % Density in g/L
C = 0.239;         % Specific heat capacity in J/g K
Ti = 350;          % Inlet temperature in K
cAi = 1;           % Inlet concentration in mol/L
UA = 5e4;          % Heat transfer coefficient in J/min K
k0 = 7.2e10;        % Pre-exponential factor in 1/min
EoverR = 8750;      % Activation energy divided by gas constant R in K
minus_deltaHr = 5e4; % Heat of reaction in J/mol

% Initial conditions
cA0 = 0.5;          % Initial concentration of A in mol/L
T0 = 350;           % Initial reactor temperature in K
init = [cA0, T0];

tspan = [0, 10];    % 10 minutes

% Simulate with cooling temperature ordered for plotting
Tc_values = [290, 300, 305]; % Cooling temperatures in K
colors = {'b', 'k', 'r'};    % Color for each cooling temperature
legend_entries = {'Tc = 290 K', 'Tc = 300 K', 'Tc = 305 K'};

% Temperature Plot
figure;
hold on;
title('Reactor Temperature Response');
xlabel('Time (minutes)');
ylabel('Temperature (K)');

% Concentration Plot
figure;
hold on;
title('Reactor Concentration of A');
xlabel('Time (minutes)');
ylabel('Concentration of A (mol/L)');

for i = 1:length(Tc_values)
    Tc = Tc_values(i);
    % Solve ODEs
    [t, sol] = ode15s(@(t, y) reactorODEs(t, y, V, q, rho, C, Ti, cAi, UA, Tc,
k0, EoverR, minus_deltaHr), tspan, init);

    % Extract solutions
    cA = sol(:, 1);
    T = sol(:, 2);

    % Plot temperature
    figure(1); % Select the temperature figure
    plot(t, T, 'Color', colors{i}, 'LineWidth', 2);
    % Plot concentration
    figure(2); % Select the concentration figure
```

```

        plot(t, cA, 'Color', colors{i}, 'LineWidth', 2);
    end

    % Finalize temperature plot
    figure(1);
    legend(legend_entries, 'Location', 'best');
    grid on;
    hold off;

    % Finalize concentration plot
    figure(2);
    legend(legend_entries, 'Location', 'best');
    grid on;
    hold off;

function dydt = reactorODEs(t, y, V, q, rho, C, Ti, cAi, UA, Tc, k0, EoverR,
minus_deltaHr)
    cA = y(1);
    T = y(2);
    k = k0 * exp(-EoverR / T); % Reaction rate constant in 1/min
    % ODEs
    dcAdt = q * (cAi - cA) / V - k * cA;
    dTdt = (q * rho * C * (Ti - T) + minus_deltaHr * V * k * cA + UA * (Tc - T)) / (V
* rho * C);
    dydt = [dcAdt; dTdt];
end

```